

## **Module-2**

### **Session 1**

### **(TASKS)**

**1. Write a short paragraph in your own words defining what a 'program' is and give one real-life example from any app you use daily (like Instagram, Zomato, or Paytm).**

**Ans.**

- A program is a collection of instructions written in a programming language that tells a computer or a mobile device what tasks it should perform. A program is created by programmers to solve problems or automate different activities. When a user opens an application and performs an action, the program processes the input, performs the required operations, and provides the expected output.
- In simple words, a program acts like a set of step-by-step directions that a computer follows. Without programs, computers and smartphones would not be able to perform useful tasks such as sending messages, making payments, ordering food, or watching videos.

**Real-Life Example (Paytm): -**

A good example of a program is the Paytm app. When a user opens Paytm and scans a QR code to make a payment, the program performs several tasks automatically: -

- It opens the camera for scanning.
- It reads the QR code details.
- It displays the merchant's information.
- It asks the user to enter the payment amount.
- It verifies the user's password or biometric authentication.
- It securely transfers the money.
- It displays a payment success message and sends a receipt.

All these operations are controlled by the program running behind the application.

**2. List three popular programming languages you have heard of, and for each, mention whether it is usually compiled, interpreted, or can be both.**

**<br><br><em><strong>Hint:</strong> Think of languages like Java, Python, and JavaScript. </em>**

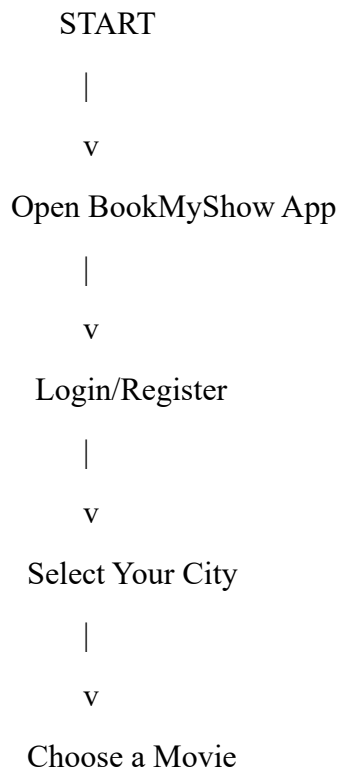
**Ans.**

## Session 1 (TASKS)

| Programming Language | Compiled / Interpreted                                                                  | Description                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Java                 | Both (Compiled and Interpreted)                                                         | Java source code is first compiled into bytecode and then interpreted or executed by the Java Virtual Machine (JVM). This makes Java platform-independent.                                    |
| Python               | Interpreted                                                                             | Python programs are generally executed by an interpreter, making it easy to write and test programs quickly. It is widely used in web development, artificial intelligence, and data science. |
| JavaScript           | Interpreted (Modern engines use Just-In-Time compilation, so it can be considered both) | JavaScript mainly runs inside web browsers and is used to create interactive websites. Modern browsers compile parts of the code during execution for better performance.                     |

**3. Draw a flowchart (on paper or using any online tool) for the process of booking a movie ticket on BookMyShow, starting from opening the app to receiving the booking confirmation.**

**Ans.**



Session 1  
(TASKS)

|                                  |    |
|----------------------------------|----|
|                                  |    |
|                                  | v  |
| Select Date and Time             |    |
|                                  |    |
|                                  | v  |
| Select Cinema Hall               |    |
|                                  |    |
|                                  | v  |
| Choose Seat(s)                   |    |
|                                  |    |
|                                  | v  |
| Confirm Seat Selection           |    |
|                                  |    |
|                                  | v  |
| Proceed to Payment               |    |
|                                  |    |
|                                  | v  |
| Select Payment Method            |    |
|                                  |    |
|                                  | v  |
| Make the Payment                 |    |
|                                  |    |
|                                  | v  |
| Payment Successful?              |    |
| /                                | \  |
| Yes                              | No |
|                                  |    |
| v                                | v  |
| Receive Booking    Retry Payment |    |
| Confirmation                     |    |

## Session 1 (TASKS)

```
|_____|  
|  
v  
END
```

**4. Write down the steps (as an algorithm, using simple numbered instructions) for ordering food on Swiggy — from opening the app to completing payment.**

**Ans.**

**Algorithm: Ordering Food on Swiggy: -**

Step 1: Start the process.

Step 2: Unlock the smartphone and open the Swiggy application.

Step 3: Log in to your account using your mobile number or email ID. If you are already logged in, proceed to the next step.

Step 4: Allow the application to access your location or manually enter the delivery address.

Step 5: Browse the list of available restaurants near your location.

Step 6: Select the restaurant from which you want to order food.

Step 7: View the menu and choose the food items you wish to purchase.

Step 8: Add the selected food items to the cart.

Step 9: Review the cart and verify the quantity, price, and selected items.

Step 10: If required, apply any available coupon code or discount offer.

Step 11: Click on the "Proceed to Checkout" button.

Step 12: Confirm the delivery address and contact details.

Step 13: Choose the preferred payment method such as UPI, Debit Card, Credit Card, Net Banking, Wallet, or Cash on Delivery.

Step 14: Complete the payment process.

Step 15: Wait for the payment confirmation from the application.

Step 16: Once the payment is successful, the order is placed with the restaurant.

Step 17: Track the order status in the app while the restaurant prepares and the delivery partner picks up the food.

Step 18: Receive the food delivery at your address.

Step 19: Verify the order and enjoy the meal.

Step 20: End the process.